VERIFICATION AND SOURCES

Mathematics, evidence, and interpretation

The scope of a proof

A theorem asserts what follows from its stated hypotheses. A conditional theorem remains a theorem under those hypotheses. Numerical evidence concerns the declared inputs, tolerances, and implementation. A physical model adds an identification of mathematical objects with quantities in nature. These roles are kept separate throughout the volume.

The free Dirac construction

The formal record covers mode algebra, completed coefficient and torus evolution, the maximal Hamiltonian, its exact first-order Fourier domain, physical self-adjointness, and the full strong generator. The centered extension proves graph-domain convergence, actual nonreal resolvent inverses and strong limits, and quadratic estimates under a finite third-order Fourier moment. The abstract spectral-functional-calculus identity remains a written deduction without a separate formal API linkage.

The comparison spaces matter

The centered operators act on the same infinite torus Fourier index space. A changing finite spatial grid additionally needs sampling, interpolation, and aliasing control. A background coupling changes the operator-domain problem. A dynamical gauge or gravitational field requires further evolution and constraint arguments.

Reading the technical appendices

The detailed companions identify exact declarations, hypotheses, source records, computations, and missing corroboration. A recorded compiler pass is not a new independent compilation. A checksum validates identity relative to a reference file, not the physical assumptions or the truth of every surrounding interpretation.

Formal and computational source identity

Pinned formal source

The preserved operator baseline is Komorebi Physics Lean v0.49.0, Checked Block 49. The unchanged Relativistic3DGeneratorBridge is extended by centered-convergence, modal-perturbation, centered-rates, and centered-resolvent modules in the KomorebiPhysics namespace. All use Lean 4.34.0-rc2 and Mathlib revision aeb72f1e9981b6d35b426b16dca7759be3e7ba06. Exact source identities belong to the reproduction record, not the mathematical chapter names.

Current and inherited compiler evidence

The current run freshly compiled twenty-six local modules and audited 209 environment declarations, including generated auxiliaries: 162 from the four new modules and 47 from the unchanged generator. Five module kernel replays and twenty-one positive or rejection controls passed. Source, object and relevant runtime identities remained stable. The sixty-four command records retain their individual outcomes. The compiler and external dependencies were pinned and reused, not rebuilt from source.

Known evidence limitations

The gauge records contain a cumulative-build log with a digest discrepancy and a Git exit-code-128 failure. Six referenced logs for two later source checkpoints are absent. Source identity checks do not turn those gaps into successful builds. The detailed gauge companion identifies the affected records. The generic aggregate Lean root also does not establish the full formal endpoint by itself. The named module and its exact import closure are the relevant target.

Numerical evidence

The preceding publication checkpoint reran five supplied suites: finite Fourier generator tests, exact Clifford and edge checks, finite adic identities, Lorentz-sector BRST parity, and the covariant backreaction benchmark. Their retained logs and input hashes preserve the energy-availability obstruction. The current formal extension does not rerun those computations or every cosmology, electrovac, and collapse simulation. External-data and historical reproduction limits remain explicit.

Physical assumptions and source conventions

Energy availability controls the matter interpretation

For the specified rapid-crossing benchmark, the unprojected spectrum exceeds the available crossing energy by about 2.63×10^{20}. A common normalization factor does not solve the coupled production equations. The cosmology calculations therefore propagate a prescribed distribution. They are conditional model results, not a demonstrated microscopic production prediction.

Gravity and arithmetic refinement

The gravitational results retain their chart, nondegeneracy, boundary, spectral-support, and numerical hypotheses. The classical master equation does not define a quantum measure or settle anomalies. Arithmetic refinement supplies written finite and infinite constructions under their stated compatibility and domain assumptions. Related formal tower interfaces are not a Lean certificate for the whole arithmetic model.

Bibliographic scope

The bibliography supplies context for the constructions and records direct comparisons where supported. It is not an exhaustive specialist literature-priority audit. The book does not infer historical priority, independent reproduction, or empirical preference from the extent of the material or from its internal checks.

Source correspondence and reproduction

The accompanying source concordance maps reading passages to the implementation notes moved out of the mathematical narrative. The editorial record preserves the provenance of the source texts, unchanged proof material, document-build results, and known verification limits. Source files, displayed figure assets, and assembly scripts allow the reading volume to be rebuilt. Compiler replay and a full numerical reproduction are separate tasks with their own dependency requirements.